

AICARLS final status

Updated 2026-09-14. Scope: verification of the existing non-API MVP. Live external-provider work remains paused by the user. No live provider requests or credential checks were performed in this continuation.

Implemented means code exists. Tested means local/deterministic/fixture checks passed. Live Verified means successful actual external execution; local fixtures are never counted as live verification.

Feature	Implemented	Tested	Live Verified	Fallback	Notes
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TXT ingestion	Yes	Passed	N/A – local	Reject invalid content	UTF-8, upload validation, source and line metadata
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PDF ingestion	Yes	Passed	N/A – local	Reject insufficient text	Actual generated two-page PDF preserves page association; scanned PDF OCR not verified
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Image/handwriting extraction	Yes	Mock passed	BLOCKED_PENDING_USER_API_SETUP	Explicit failure/uncertainty	Image validation and low-confidence handling tested; actual recognition remains pending
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Session knowledge and SQLite persistence	Yes	Passed	N/A – local	None	Reopened store retains knowledge, mastery and roadmap
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| Isolated session **RAG** | Yes | Passed | **N/A** – local | No-evidence response | **FTS** retrieval, original-source preference, cross-session isolation; no production authentication claim |

| Tutor and citation association | Yes | Mock/local passed | **BLOCKED_PENDING_USER_API_SETUP** | Labeled source excerpts | Exact chunk citations validated; invented citations rejected; live generated answers pending |

| Topic source adapters | Yes | Mock passed | Not verified **in this** run | Labeled generated material | No **public**-source network requests made; generated lesson provider pending **API** setup |

| Assessment generation | Yes | Passed | **N/A** – deterministic | Reject insufficient evidence | Five MCQs and one short answer; source/concept mappings and hidden answer keys checked |

| Scoring and knowledge gaps | Yes | Passed | **N/A** – local | Keyword rubric | Weighted concept scores and thresholds tested; short-answer scoring is not a reasoning-quality evaluator |

| Personalized roadmap and retest | Yes | Passed | **N/A** – local | Requires assessed concepts | Actual gaps drive roadmap; retest updates assessed concepts without erasing others |

| Closed learning loop through local FastAPI | Yes | Passed | **N/A** – fixture | Mock tutor | **TXT** input → knowledge → **RAG** → cited tutor → assessment → scoring → gaps → persisted roadmap **in** one test; completion gate explicitly simulated, not a live video lesson |

| FastAPI startup and route contracts | Yes | Passed | **N/A** – local | Safe **404/422** errors | TestClient startup, session/upload/assessment/chat/gaps/roadmap responses; running proxy returns expected **404** **for** absent session |

| LearningInput / LearningLoop frontend | Yes | Build and scoped lint passed | **N/A** – local | Displays backend errors | Integrated Workspace and SessionContext checked; homepage **HTTP 200** and backend proxy verified. No browser interaction walkthrough performed |

| Deterministic semantic video | Yes | Passed once | N/A – local | No template/stub/render fallback allowed | Cached Piper, Manim, FFmpeg; first render succeeded, non-silent audio, final duration 11.01 seconds; uniform word alignment |

| Live lesson planning/narration and targeted video generation | Yes | Failure handling tested | BLOCKED_PENDING_USER_API_SETUP | Failed stage persisted | Full live provider-driven lesson remains pending; deterministic component video does not establish live end-to-end success |

| Provider abstraction | Yes | Mock regressions passed | BLOCKED_PENDING_USER_API_SETUP | Explicit errors and labeled fallback paths | No auth investigation, credentials inspection, provider probes or live LLM calls this run |

| Docker / Compose environment | Yes | Existing services running | N/A – local | None | Backend port 5002, frontend 3002. Baseline build passed previously; fresh image rebuild skipped this run. Current containers contain previously installed updated dependencies; fresh-image reproducibility is not newly verified |

Verification evidence

- Existing targeted learning-loop module: **28 passed**.
- Added local closed-loop API/persistence integration test: **1 passed**.
- Frontend production build: **passed**, 42 modules.
- Scoped ESLint: **passed** for LearningInput, LearningLoop, VideoPlayer, Workspace and SessionContext. Historical lint cleanup deferred.
- Deterministic semantic video regression: **passed once**, Python socket connections blocked and gTTS disabled for this invocation;

cached voice files checked before rendering.

- Final full backend suite: ****58 passed****, run once after targeted checks. Three existing dependency deprecation warnings; no failures.

- Existing Compose services: both running. Frontend `` returned **200**; proxied missing learning-session endpoint returned expected **404**.

Video:

``backend/data/renderers/verified_work_energy_e283bfd5/work_energy.mp4``

Video report:

``backend/data/renderers/verified_work_energy_e283bfd5/verification.json``

Changes in this continuation

Added a single combined **API** closed-loop regression test covering persisted results and isolation. Completed the interrupted VideoPlayer drawCanvas hoist and verified the integrated frontend. No production learning behavior was weakened for fixture success.

Pending user setup

``BLOCKED_PENDING_USER_API_SETUP``: live generated tutoring, image recognition, provider-based lesson planning/narration and targeted visual relearning require configured external-provider access. These

branches remain paused. Their mock tests demonstrate local contracts only. No further **API** work should run until the user instructs it.

The non-**API** verification scope is complete. Optional features, architecture changes, historical lint and repeated builds/renders are deferred.